

Chaos Language Algorithm Experiments

A Comprehensive Evidence-Bounded Report

Chaos Language Algorithm / OmegaSim projects

Evidence cutoff: July 15, 2026

Abstract

This report consolidates every durable Chaos Language Algorithm (CLA) experiment found in the project and repository records through July 15, 2026, with emphasis on the newer high-dimensional work absent from earlier reports. CLA currently provides exact-reconstructing symbolic grammar induction, low-cardinality and fixed symbolization, an approximate two-part compression proxy, shuffled-surrogate comparisons, and held-out n-gram diagnostics. Across the evidence, exact reconstruction is reliable, raw rectangular partitions fail in high dimension, and low-cardinality microstates restore recurrence. However, positive compression margins are not unique to VAMP embeddings, and a leakage-free rank-conditioned comparison rejected further linear-VAMP tuning: VAMP beat matched PCA controls on both primary metrics in zero of three trajectories at both alphabet sizes. The report therefore separates positive engineering results, detector failures, null comparisons, and hypotheses; it does not claim a validated chaos classifier or calibrated CLA likelihood.

1 Scope, sources, and epistemic discipline

The canonical code is the `chaoslang` repository on branch `agent/hd-embedding-cla`; the report build uses commit `974af31` plus this document. OmegaSim detector integration uses branch `agent/cla-detector-prereg`, commit `5a6002b`. Durable run records are under `projects/chaos-language-algorithm/experiments`, repository `experiments`, and `projects/omegasim/experiments`.

Observed: denotes direct code, artifact, or recorded-run evidence. **Inferred:** denotes interpretation that follows from those observations. Proxy quantities retain historical “bits” field names where necessary, but the current scorer is not a complete calibrated code. Held-out perplexity is from a smoothed n-gram model over symbols, not from a predictive CLA grammar.

2 Detector pipeline and what each metric establishes

The experimental pipeline is

$$x_{1:n} \rightarrow \text{symbolizer} \rightarrow s_{1:n} \rightarrow \text{CLA grammar and exact parse.}$$

CLA greedily proposes repeated chunks and contextual categories while requiring the final grammar/parse to expand exactly to the original symbol stream.

Measurement	What it directly checks	What it does not establish
Exact reconstruction	Grammar and parse losslessly reproduce the observed symbols	Generating partition, chaos, prediction, or useful abstraction

Measurement	What it directly checks	What it does not establish
Compression proxy	Current grammar representation is shorter/longer than an empirical-symbol baseline under the implemented proxy	A calibrated MDL code, likelihood, or significance level
Real-minus-surrogate margin	Temporal order helps relative to a specified shuffle/block surrogate	Uniqueness to the proposed embedding or dynamics
Held-out n-gram loss	Symbol stream has train-to-test finite-order predictive regularity	Predictive performance of the CLA grammar itself
Embedding singular values	Numerical modes returned by a kinetic estimator	Validated slow processes when modes are near one or conditioning is weak

3 Baseline attractor and scaling experiments

The first lane used an equal-width rectangular partition (M1) and a small greedy CLA budget. All recorded runs reconstructed exactly. Table 2 consolidates the baseline records.

System	Dim.	Samples	Unique symbols	Rules	Proxy score
Mackey–Glass	1	256	4	6	44.4
Logistic map	1	256	4	8	113.2
Lorenz–63	3	256	21	8	187.2
Rossler	3	256	24	8	189.2
Lorenz–96	5	256	56	8	234.2
Lorenz–96	8	256	78	8	229.2
Lorenz–96	12	256	107	6	241.4
Lorenz–96	16	256	121	3	248.7
Lorenz–96	20	256	127	6	251.4
Lorenz–96	24	256	148	5	251.5
Lorenz–96 trie/JS	20	1024	510	8	1021.2

Table 2: Recorded M1 experiments. Scores are proxy units, not calibrated bits.

Observed: In the strongest high-dimensional M1 record, 1024 samples of 20-dimensional Lorenz–96 yielded 510 unique compound symbols, eight accepted chunk rules, no category, score 1021.2, and 2.55 seconds fit time. Provenance: 20260709T192728Z-lorenz96-1024-dim20-suffix-trie at commit b344aa3c. The score stayed close to raw sequence length.

Inferred: The main failure occurred before grammar induction: the Cartesian partition produced an alphabet too sparse for robust recurrence. The run is not evidence that Lorenz–96 lacks symbolic grammar. It motivated choosing alphabet cardinality directly with microstates.

4 Lifted Lorenz–63 Phase-1 experiment

The first adaptive high-dimensional experiment lifted Lorenz–63 into $\mathbb{R}^{256}$ with small deterministic noise, used a dependency-free kinetic map to dimension three, clustered microstates, and compared with raw M1.

Observed: In the one-seed diagnostic, adaptive microstates achieved a real-minus-shuffled compression proxy of 39.33 versus 2.10 for M1. Held-out n-gram perplexity was 3.42 versus 433.18. Exact reconstruction held. Provenance: 20260712T200000Z-lorenz63-lift256-phase1.

Inferred: Low-cardinality symbolization decisively repaired the M1 alphabet explosion on this fixture. The run did not isolate the kinetic embedding from clustering, used one trajectory, and did not compare against direct raw-space k-means. It therefore supported the pipeline seam, not VAMP superiority.

5 Lifted Lorenz–63 multi-seed VAMP sweep

The next preregistered sweep used three seeds, a 192-point retained trajectory, a noisy $\mathbb{R}^{256}$ lift, deeptime VAMP dimensions $\{3, 5\}$, alphabet sizes $\{8, 16, 24\}$, and lags $\{1, 3\}$. It produced 36 VAMP rows and nine controls.

Method/aggregate	Rows	Mean proxy margin	Mean n-gram perplexity
deeptime VAMP overall	36	8.085	11.628
direct raw xyz k-means	3	12.200	3.697
dependency-free reference	3	15.167	4.151
raw compound M1	3	1.867	257.769

Table 3: Aggregate multi-seed results; proxy margins are not calibrated MDL.

Observed: All 36 VAMP configurations had positive real-minus-shuffled proxy margins, but direct xyz k-means and the dependency-free reference were as good or better on the aggregate metrics. No categories emerged. The recorded near-unit singular values exposed a diagnostics bug: finite-only validation had failed to flag invalid or implausible implied timescales. The writer was repaired to preserve raw values and fail closed. Provenance: 20260712T200200Z-lorenz63-r256-deeptime-multiseed, commits bee5965 and 64e9dd6.

Inferred: Positive shuffled-surrogate margins were a property of many low-cardinality streams, not evidence uniquely favoring VAMP. The correct next comparison required leakage-free train-only transforms and alphabet-matched PCA/raw controls.

6 Leakage-free rank-conditioned VAMP/TICA test

The decisive follow-up used three independent 2048-point Lorenz–63 trajectories, train-only noisy $\mathbb{R}^{256}$ lifts, 60/20/20 blocked splits with 16-point lag gaps, PCA ranks $\{3, 10\}$, VAMP/TICA dimensions $\{2, 3\}$, lags $\{1, 4, 16\}$, $k = \{8, 16\}$, and 20 circular block surrogates at block lengths 4 and 16. Selection occurred on blocked validation VAMP-2 and test was evaluated once.

Seed	k	VAMP loss	best PCA loss	VAMP margin	best PCA margin
101	8	0.434	0.420	5.20	3.65
202	8	0.482	0.423	5.20	5.55
303	8	0.615	0.543	13.00	7.30
101	16	0.731	0.710	4.30	2.60
202	16	0.771	0.727	4.25	3.05
303	16	1.050	0.952	3.90	9.00

Table 4: Test loss in bits/symbol; margin is the minimum proxy margin across block lengths. Lower loss and higher margin are better.

Observed: VAMP beat the best matched PCA control on both metrics in zero of three trajectories for $k = 8$ and zero of three for $k = 16$, versus the preregistered requirement of at least

two of three. VAMP test loss was worse in every seed/alphabet stratum. All 72 estimator candidates were numerically valid; selected modes were near-unit, so they were not treated as standalone kinetic evidence. Runtime was 163 seconds and the full suite passed 115 tests. Provenance: 20260713T153839Z-lorenz63-rank-conditioned-vamp-tica, commit 4423f8f.

Inferred: Linear VAMP development correctly stopped. This bounded null result does not reject nonlinear embeddings, other dynamics, or CLA; it rejects the claim that the tested linear VAMP path improved on matched PCA for this lift.

7 OmegaSim restart under a fresh detector preregistration

Ben approved restarting OmegaSim on July 15 under a new preregistration. The frozen design evaluates A6 gain 5.0, coupling {0.35, 0.60}, delay {0, 3}, controls {appraisal, linear, shuffled}, five seeds, and separate 4D, 8D, and 20D feature strata. A deterministic $k = 8$ microstate model is fit on the first 60 percent of each trajectory and frozen before evaluation. Each row uses five marginal-preserving symbol surrogates.

The row-level detector requires exact reconstruction, at least four occupied states, positive proxy margin, and held-out order-2 loss at least 0.10 bits/symbol below the surrogate mean. Promotion additionally requires appraisal to beat both matched dynamics controls on both metrics in at least four of five seeds, without pooling dimensionality strata. Full protocol: `omegasim/docs/cla_detector_preregistration_20260715.md`.

Observed: The local CPU run completed all 180 rows in 31 minutes 26 seconds. All row-level software invariants passed. Exactly one of the twelve frozen tuple/stratum promotion cells met the four-of-five matched-seed criterion: coupling 0.60, delay 3, and the 8D role stratum. Appraisal beat both linear and shuffled controls on compression margin and held-out loss in four seeds; seed 29 missed because its compression margin (118.6 proxy bits) was below the linear control (132.2), although its held-out loss was lower. No other cell reached four wins; the next highest cells reached three.

Inferred: This is bounded positive evidence that the frozen CLA feature path distinguishes appraisal dynamics from the two exact-tuple controls in one predeclared regime. It is not evidence that the dynamics form a strange attractor, that roles are semantic grammar categories, or that CLA is a calibrated chaos detector. The singled-out cell now requires an untouched replication and the still-required Mackey–Glass/Lorenz–96 calibration battery. Immutable artifacts have SHA-256 2539b8381b8e85317a7ca4ae53643483153c8f3535dabbef282abec880902eb4 (JSON) and 5dd819af1b76b40811712b8452adab8c5bcb57f1066847bf56cb779a7bd6e729 (CSV).

8 Implemented but not yet decisive experimental lanes

- Train-fitted deterministic k-means symbolization is now a first-class leakage-safe object (commit 974af31); functional tests establish no test-data refit, not scientific superiority.
- Frozen-suffix coding and deterministic model encodings were implemented on a separate branch (commits 5be45e5, ce0e476) with 140 full tests passing. The preregistered held-out benchmark has not yet produced a durable result, so calibrated predictive coding remains open.
- PCCA+ soft memberships exist in the optional backend, but automatic category seeding and a decisive category recovery experiment remain pending.
- The bounded suffix trie has functional/node-bound tests and one 2.55 s Lorenz–96 run, but no matched wall-time/peak-memory comparison against the old brute-force miner.

9 Current implementation validation

Observed: On July 15, the command `PYTHONPATH=srcpython3-munittestdiscover-stests-v` completed 126 tests in 47.394 seconds with status OK on the report branch. This includes the optional deeptime-backed tests available in the local environment and the new train-fitted k-means no-refit regressions. A passing software suite establishes the declared invariants and fixture behavior; it does not convert the null scientific comparisons above into positive evidence.

10 Conclusions and claim ledger

Claim	Status	Evidence / qualification
CLA exactly reconstructs tested symbolic streams	Supported	All durable baseline and high-dimensional runs plus regression tests
Raw rectangular M1 scales poorly in high dimension	Supported for tested fixtures	510 unique symbols in 1024 by 20D; near-raw proxy score
Low-cardinality microstates restore recurrence	Supported	Lifted one-seed and multi-seed runs; direct controls also strong
Positive proxy margin uniquely indicates grammar-bearing embedding	Rejected	Direct raw/reference controls were equally or more positive
Linear VAMP improves matched PCA on lifted Lorenz-63	Rejected for frozen protocol	Zero of three joint wins at both k values
CLA is a validated chaos detector	Not established	No calibrated operating curve, likelihood, or matched broad benchmark battery
CLA categories recover meaningful attractor lobes	Not established	No categories in the main high-dimensional sweeps
OmegaSim appraisal dynamics differ under the frozen CLA proxy in one regime	Bounded support	One of twelve cells promoted: coupling 0.60, delay 3, roles8, with matched wins in four of five seeds; untouched replication required

The central scientific lesson is methodological: symbolization and controls dominate interpretation. Exact reconstruction is necessary but easy; positive compression relative to a weak shuffle is insufficient; and a useful embedding must beat alphabet-matched, train-only alternatives on held-out prediction and temporal-grammar diagnostics simultaneously. The current evidence supports CLA as an inspectable experimental grammar toolkit. It does not yet support CLA as a calibrated strange-attractor classifier.

Reproduction pointers

Each project-level experiment directory contains `RUN.md`, `command.sh`, `environment/git` captures, status, logs, raw JSON/CSV, and checksums where available. The decisive records are:

- 20260709T192728Z-lorenz96-1024-dim20-suffix-trie
- 20260712T200000Z-lorenz63-lift256-phase1
- 20260712T200200Z-lorenz63-r256-deeptime-multiseed

- 20260713T153839Z-lorenz63-rank-conditioned-vamp-tica
- 20260715T153049Z-cla-detector-preregistered (OmegaSim)